



The DC Spears

Basic cloud infrastructure simulating a high performance environment using IBM & Digital Ocean

Authors: Anthony Joseph, Maanda Mulaudzi, Tshifhiwa Netshifhehe, Thabo Bosvark, Allen Van Dieman(mentor), Mabatho Hashatsi (CSIR mentor), Nyameko Lisa(CSS Organisers)

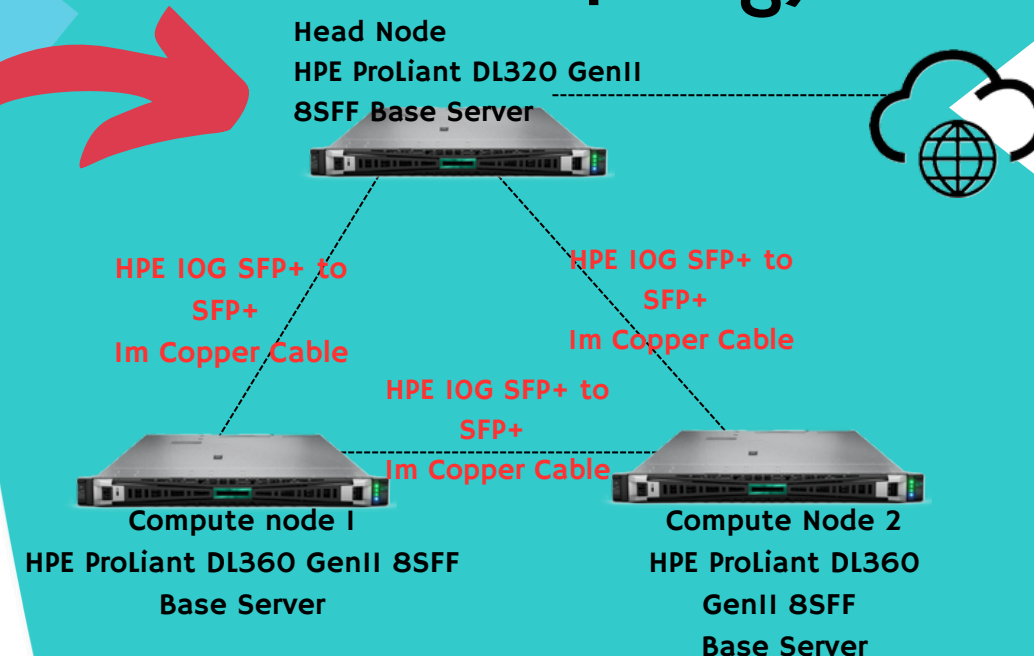


1. Introduction

The SCC required teams to design a multi-node HPC cluster, selecting appropriate CPUs, RAM, power budgets, and other server components.

The DC Spears worked on a cloud-deployment project simulating a high-performance environment using IBM Cloud and DigitalOcean. The project focuses on automating instance creation and management through Terraform, Ansible, and CI/CD pipelines

Network Topology



2. Cluster design

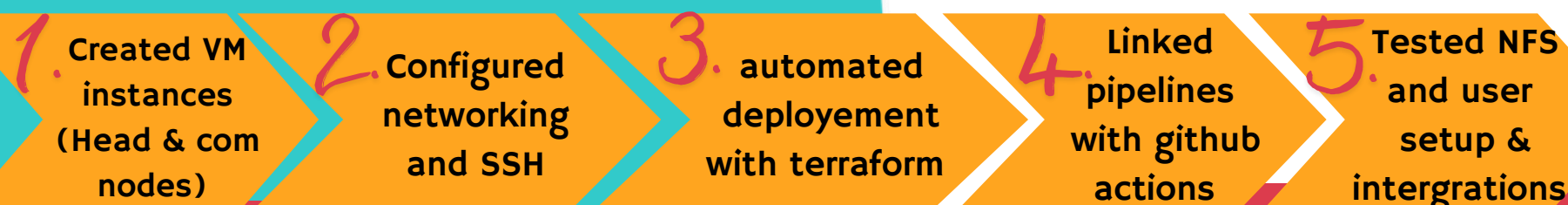
- Rocky Linux based
- Intel Xeon CPUs
- 32 core head node & 64 cores compute nodes
- 800W power supply
- 2.4TB head node storage +300GB compute nodes storage



3. Cluster considerations

- Runs AmberMD, ASCOT5, DFTB+ & Mathworks applications
- Multicore CPUs per node (64 cores com nodes) for parallel execution (Matlab) and matrix operations for quantum simulations
- Fast interconnects for MPI scaling of fusion orbit simulations
- With 160 cores across the cluster, the system is well suited for benchmarks and AI applications.

4. Process flow



5. Software Tools

- Terraform— Infrastructure as Code
- Ansible - Configuration Management
- GitHub Actions & CircleCI
- MobaXTerm - SSH access

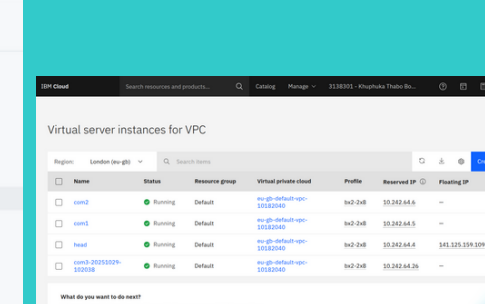
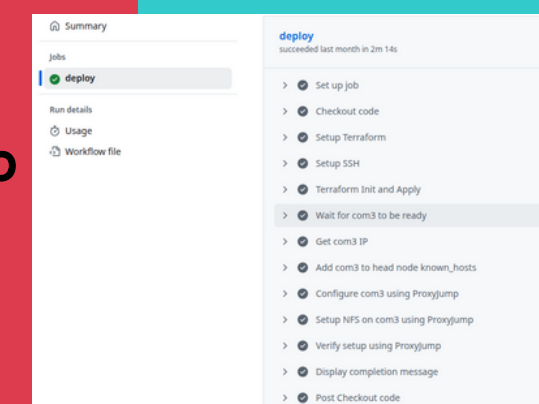


6. CI/CD Pipelines

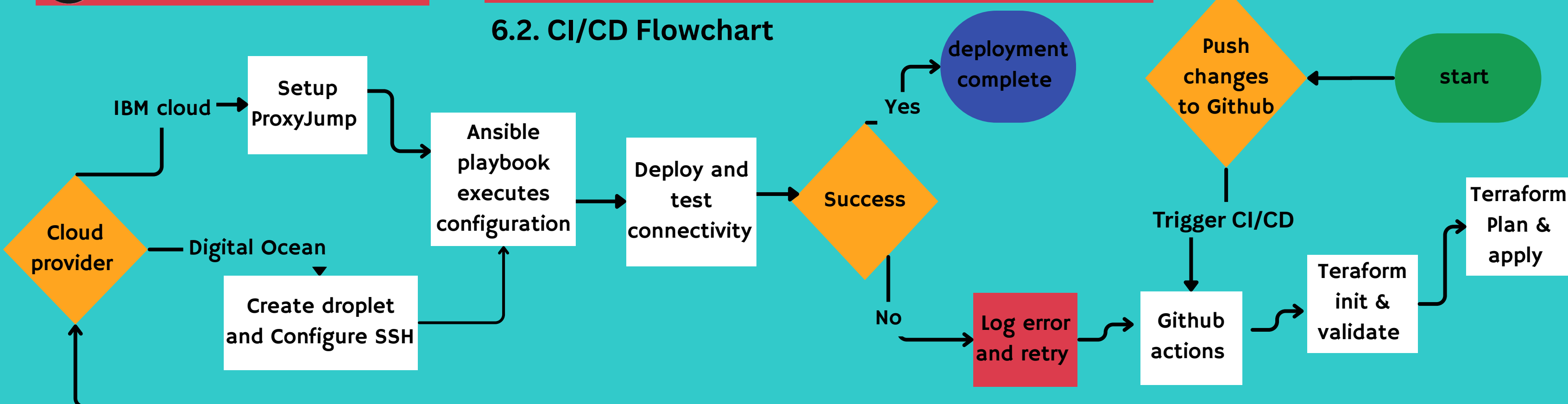
We automated the deployment and configuration of the HPC cluster using github Actions and ansible.

6.1. GitHub repository Setup

- separate GitHub repos for IBM cloud and DigitalOcean
- Repository secrets and private keys for cluster access added in Github secrets



6.2. CI/CD Flowchart



7. Findings

- The automated deployment pipeline cut setup from hours to just minutes.
- Tasks like VM creation, node configuration and NFS setup, once manual and time-consuming, are now completed with a few clicks.
- see our Github Repo for more

8. Challenges

Navigating complex hardware choices and optimizing benchmarks was daunting for a group with limited prior experience. We combined our strength, learned rapidly and managed to push through.

